O-LEVEL PHYSICS • PAPER 3 PRACTICAL

Mock Question Pack

Syllabus 6091 • ~4-Hour Teaching Set

Tutor edition — questions, diagrams & full answer key

How to use this pack. Eight questions arranged as two mock sections plus a focused planning & evaluation block — roughly four hours of teaching once you talk through technique.

- Each question is tagged by skill area (**P / MMO / PDO / ACE**) so you can target weaknesses.
- Blue dashed boxes are **teaching notes** — what to emphasise, common pitfalls.
- Diagrams are drawn so the tutee can read scales and take off values.
- The **full answer key with worked marks** is at the very end (from page 13).

Suggested pacing is on the next page.

Written to the SEAB Singapore–Cambridge O-Level Physics syllabus 6091 (first examined 2026): Paper 3 is 40 marks / 20%, two 55-minute sections, assessing Planning, MMO, PDO and ACE. Figures and data are illustrative teaching material, not past-paper reproductions. Confirm apparatus details against current SEAB documents.

Teaching plan — four hours

A workable order. Teach the technique first, then let the tutee attempt each question cold before you mark it together against the key.

0:00	Warm-up: how Paper 3 is marked (P/MMO/PDO/ACE), reading instruments to full precision, the consistency rule for tables.	**20 min**
0:20	**Q1** Density of an irregular solid — MMO + PDO. Focus on displacement method & parallax.	**25 min**
0:45	**Q2** Simple pendulum → g — PDO + ACE. Focus on plotting T^2 vs L and gradients.	**35 min**
1:20	**Q3** Principle of moments — MMO + ACE. Break / stretch after.	**25 min**
1:45	**Q4** Resistance of a wire — MMO + PDO + ACE. The classic circuit question.	**35 min**
2:20	**Q5** Refraction through a glass block — MMO + PDO. Ray tracing & protractor work.	**30 min**
2:50	**Q6** Cooling / specific heat — PDO + ACE. Reading a cooling curve, heat-loss error.	**30 min**
3:20	**Q7** Planning task — the 15%. Model the five-part skeleton out loud.	**25 min**
3:45	**Q8** Errors & evaluation drill — short-answer ACE. Wrap-up & checklist.	**15 min**

Marking together

Don't just give the answer — have the tutee find the mark they missed. Most lost marks in Paper 3 are technique (units in headings, sensible scales, gradient triangle too small, vague improvements), not physics.

Mock Section A

Questions 1–3. In a real exam these share a 55-minute apparatus window worth 20 marks. Here they are spaced out for teaching.

Question 1 DENSITY OF AN IRREGULAR SOLID [7 marks]

MMO PDO

A student is given a small irregular stone, a 100 cm³ measuring cylinder, water, an electronic balance and a length of thread. She is asked to find the density of the stone.

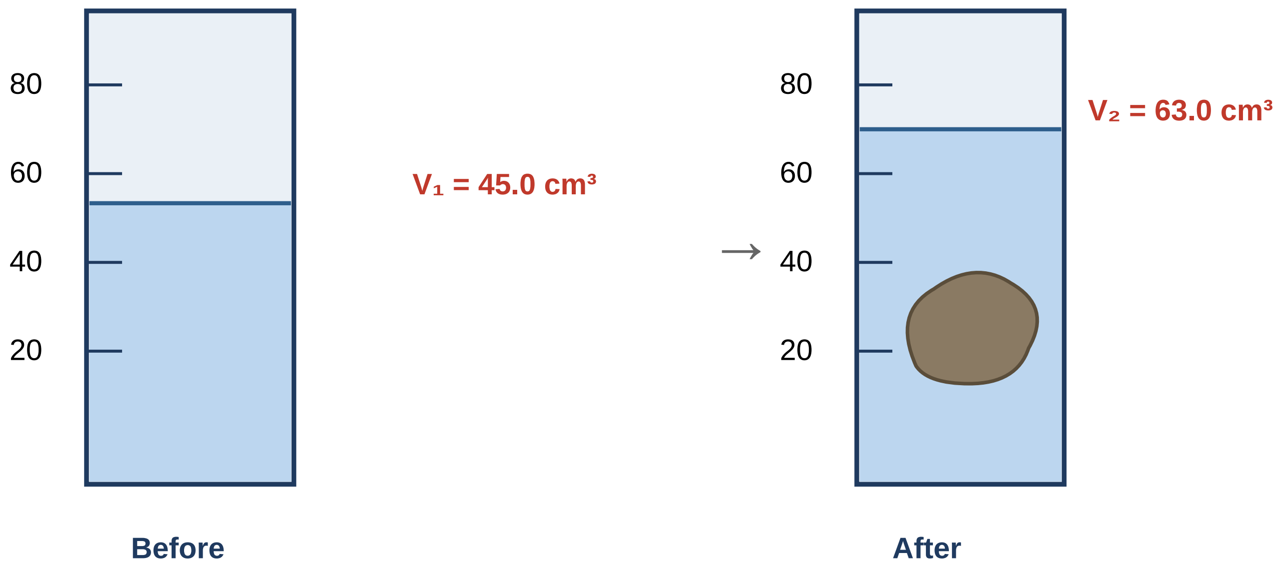

Fig. 1.1 — water level before and after the stone is lowered in. Balance reads mass of stone = 47.25 g.

1. State how the student should place her eye when reading the water level, and explain why. **[2]**
2. Using the readings in Fig. 1.1, determine the volume of the stone. **[1]**
3. The balance reads the mass of the stone as 47.25 g. Calculate the density of the stone, giving your answer to an appropriate number of significant figures and a unit. **[2]**
4. Suggest one reason the measured volume might be slightly too large, and how to avoid it. **[2]**

Teach here: the displacement method and parallax. Ask the tutee *why* we read the bottom of the meniscus at eye level before giving the rule. For (c), drill the significant-figure logic: raw data has 3 s.f., so the answer has 3 s.f.

Question 2 SIMPLE PENDULUM → ACCELERATION OF FREE FALL **[11 marks]**

PDO ACE

A student investigates how the period T of a simple pendulum depends on its length L. For each length she times 20 complete oscillations and calculates the period. Her results are shown in Table 2.1.

L / m	Time for 20 swings / s	T / s	T^2 / s^2
0.200	17.9	0.895	0.80
0.400	25.4	1.270	1.61
0.600	31.0	1.550	2.40
0.800	35.8	1.790	3.20
1.000	40.1	2.005	4.02

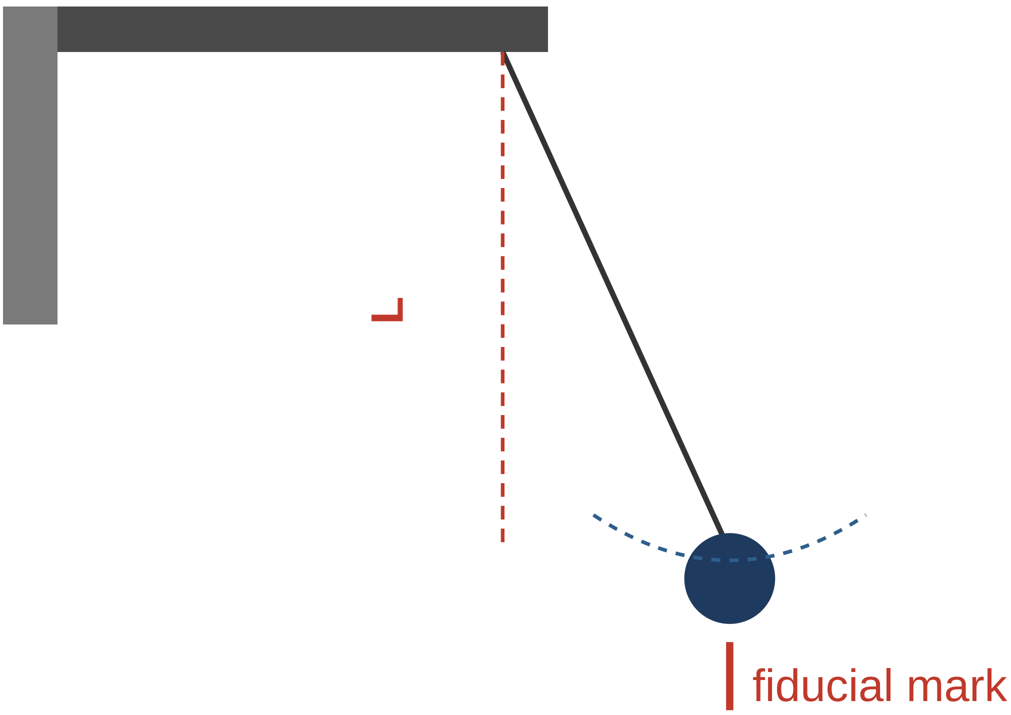

Fig. 2.1 — the pendulum. Timing is started as the bob passes the fiducial mark at the centre.

1. Explain why the student times 20 oscillations rather than a single one. **[2]**
2. On the grid (Fig. 2.2), plot a graph of T^2 (y-axis) against L (x-axis) and draw the best-fit line. **[3]**
3. Determine the gradient of your line, showing your triangle clearly. **[2]**
4. The period is related to length by $T^2 = (4\pi^2/g)\,L$. Use your gradient to find a value for g. **[2]**
5. State one reason your value of g might differ slightly from 9.81 m/s². **[1]**

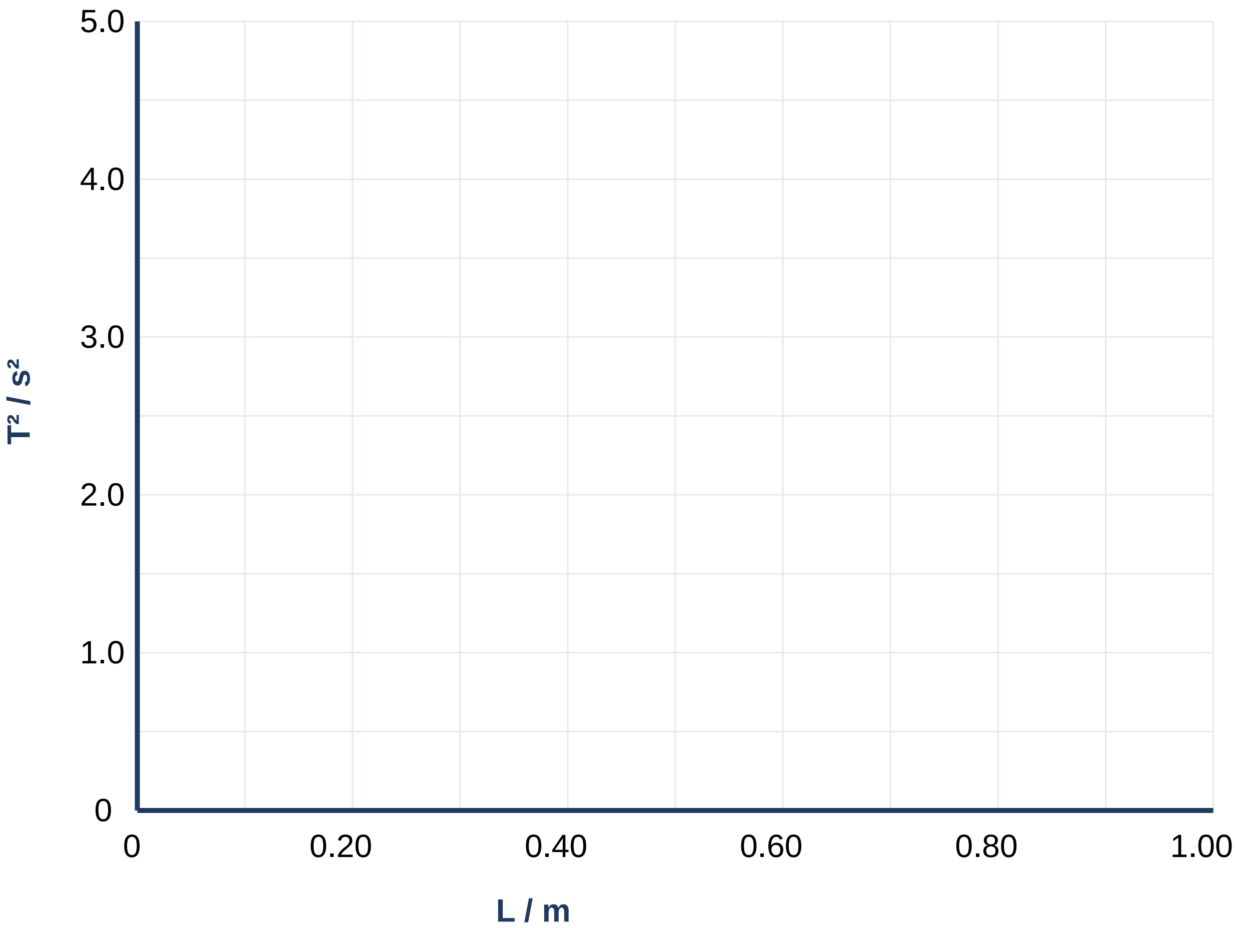

Fig. 2.2 — grid for plotting. (Each labelled square = 0.20 m on x, 1.0 s^2 on y.)

Teach here: this is the money question. (b) drill: label axes with unit, points as small crosses, best-fit with balance above/below. (c) insist the gradient triangle spans > half the line and values are read *off the line*. (d) rearrange $g = 4\pi^2/\text{gradient}$ — watch they don't invert it.

Question 3 PRINCIPLE OF MOMENTS **[8 marks]**

MMO ACE

A uniform metre rule is balanced on a knife-edge pivot at its centre. A 100 g mass is hung at the 15.0 cm mark. A mass *M* is moved along the other side until the rule balances horizontally at the 80.0 cm mark.

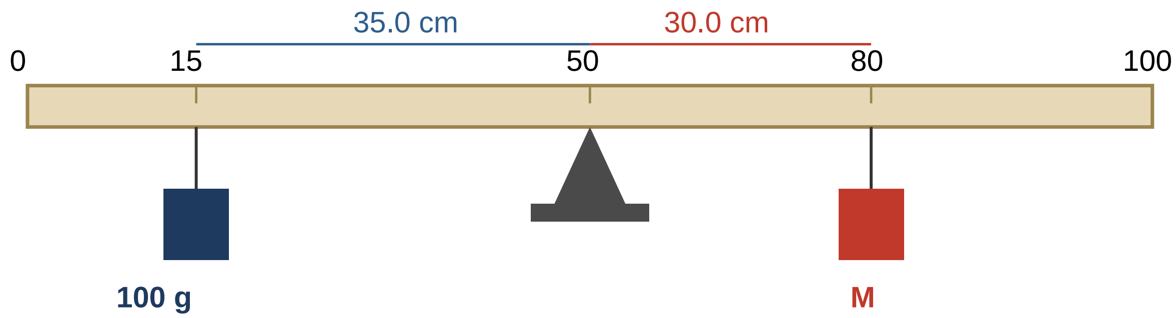

Fig. 3.1 — the balanced metre rule (drawn not to scale).

1. State the principle of moments. **[1]**
2. Calculate the perpendicular distance of each mass from the pivot. **[1]**
3. By taking moments about the pivot, determine the value of the mass *M*. **[3]**
4. The rule is uniform and pivoted at its centre. Explain why the weight of the rule does not appear in your calculation. **[1]**
5. Suggest one precaution the student should take to obtain an accurate balance point, and explain how it helps. **[2]**

> **Teach here:** moment = force × perpendicular distance, and that in a balance you can equate *mass*×distance on each side because g cancels. Part (d) is a favourite — centre of gravity acts at the pivot so its moment is zero.

Mock Section B

Questions 4–6. Longer, integrated experiments — the type that carries the full 20-mark Section B question.

Question 4 RESISTANCE OF A WIRE [12 marks]

MMO PDO ACE

A student investigates how the resistance *R* of a length of nichrome wire depends on its length *L*. The circuit in Fig. 4.1 is used. For each length, the voltmeter reading *V* and ammeter reading *I* are recorded.

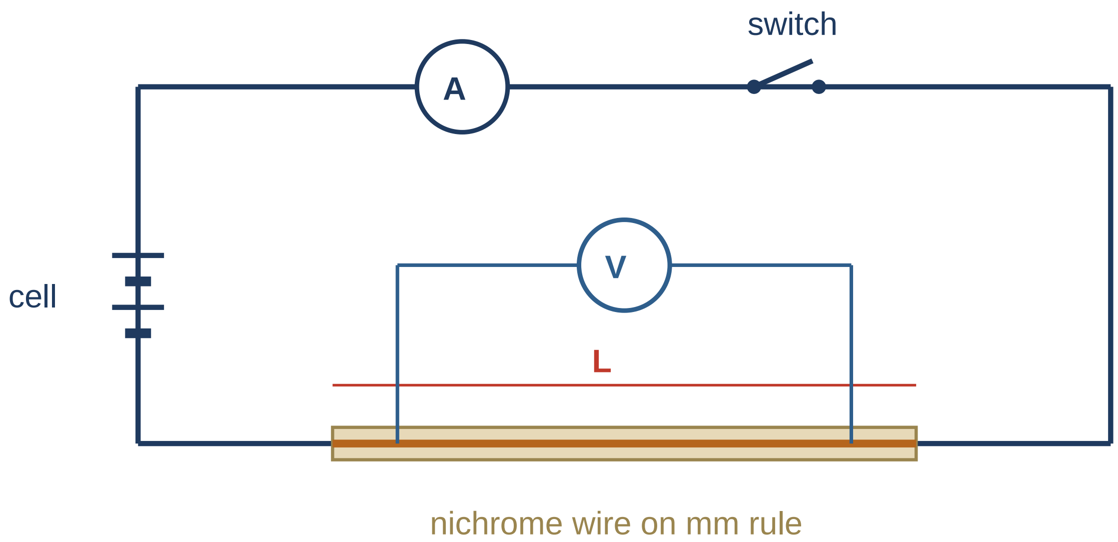

Fig. 4.1 — circuit: ammeter in series, voltmeter across the length of nichrome wire.

L / cm	V / V	I / A	R = V/I / Ω
20.0	0.42	0.60	
40.0	0.84	0.60	
60.0	1.26	0.60	
80.0	1.68	0.60	
100.0	2.10	0.60	

1. Copy and complete the final column of the table by calculating *R* for each length. **[2]**
2. State the relationship between *R* and *L* shown by your values, and explain how the numbers support it. **[2]**
3. Sketch the shape of the graph of *R* against *L* you would expect, labelling the axes. **[2]**
4. The student is told to switch off the circuit between readings. Explain why, referring to the effect on the wire. **[2]**
5. The ammeter reading stays at 0.60 A for every length. Suggest what the student must have done to keep the current constant, and name the component used. **[2]**
6. Identify one source of error in measuring *L* and state how to reduce it. **[2]**

Teach here: R = V/I per row, then spot R ∝ L (doubling L doubles R). Part (d) heating → resistance rises → systematic error. Part (e) is subtle: a rheostat re-adjusted each time keeps I fixed. Good place to discuss the difference between the two meters' positions.

Question 5 REFRACTION THROUGH A GLASS BLOCK **[9 marks]**

MMO PDO

A student traces a ray of light passing through a rectangular glass block. She marks the incident ray with two pins (P_1, P_2) and the emergent ray with two more (P_3, P_4), then removes the block and draws the ray path shown in Fig. 5.1.

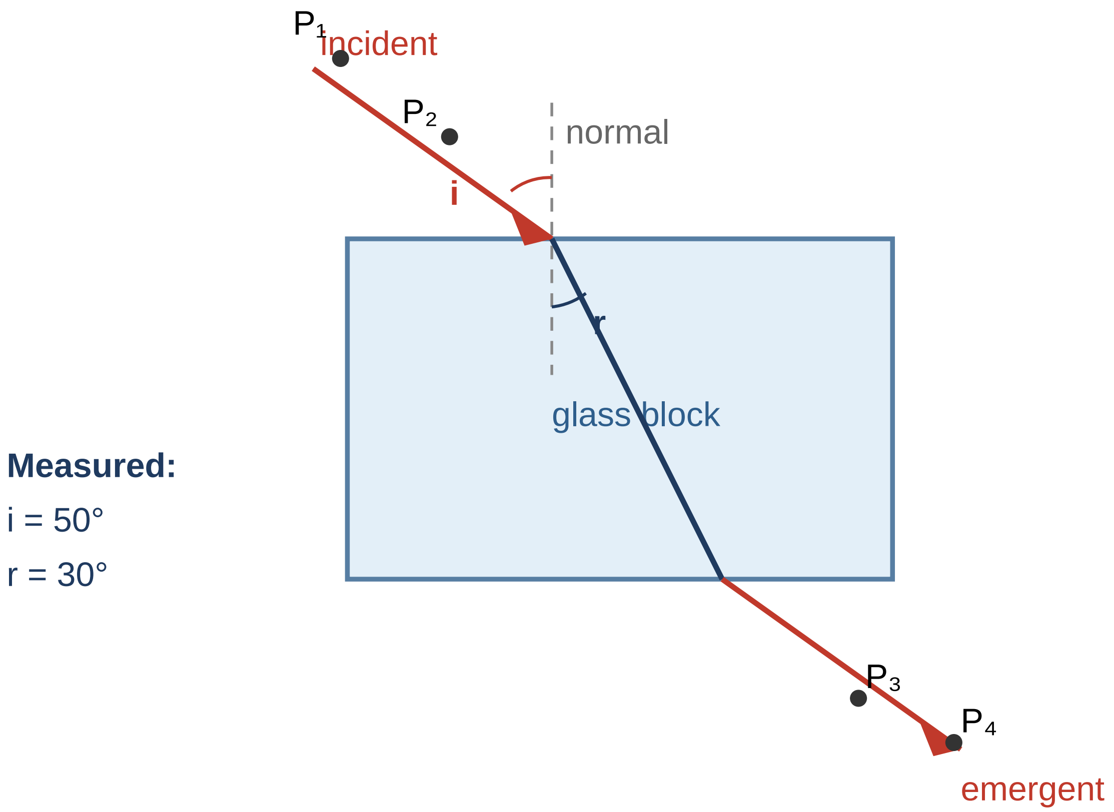

Fig. 5.1 — traced ray path. Angles measured from the normal: i = 50°, r = 30°.

1. Describe how the student should position the two pins P_1 and P_2 so that the incident ray is accurate. **[2]**
2. Explain why the ray bends *towards* the normal as it enters the glass. **[2]**
3. Using the measured angles, calculate the refractive index of the glass. (Use sin 50° = 0.766, sin 30° = 0.500.) **[2]**
4. State the number of significant figures appropriate for your answer and why. **[1]**
5. Suggest two ways the student could reduce error when measuring the angles. **[2]**

Teach here: pins vertical and well separated (long baseline = smaller angular error). n = sin i / sin r. Emphasise measuring angles *from the normal*, not the surface — a very common slip.

Question 6 COOLING CURVE & HEAT LOSS **[10 marks]**

PDO **ACE**

Hot water is placed in a beaker and its temperature recorded every minute as it cools. The results are plotted in Fig. 6.1.

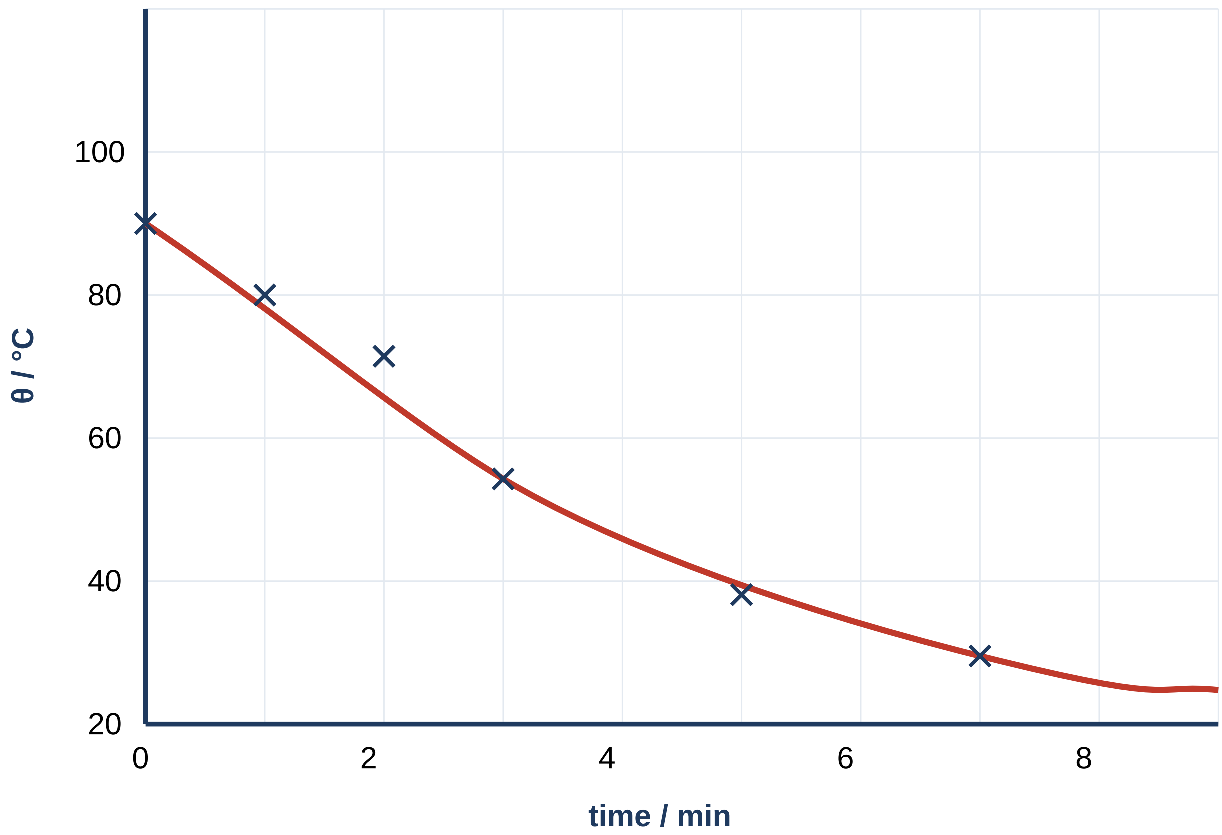

Fig. 6.1 — temperature of the cooling water against time.

1. Use the graph to state the temperature of the water after 4.0 minutes. **[1]**
2. Describe how the *rate* of cooling changes over the 8 minutes, and explain why in terms of temperature difference with the surroundings. **[3]**
3. The student wants to reduce the rate of heat loss so the water stays hot longer. Suggest two changes to the apparatus and explain how each helps. **[4]**
4. Explain why the temperature levels off rather than falling to 0 °C. **[2]**

Teach here: reading a curve (drop a line from t = 4). The rate = steepness of the curve, which decreases as the water approaches room temperature (smaller temperature difference → slower transfer). Levels off at room temperature. Great question for the "heat loss to surroundings" error that recurs in thermal experiments.

Planning & Evaluation

Questions 7–8. Planning is worth 15% of Paper 3, and evaluation marks are scattered through every question. These two build the habits directly.

Question 7 PLANNING TASK [8 marks]

PLANNING

A spring extends when a load is hung from it. A student wants to investigate how the extension of a spring depends on the load applied, and to find the load at which the spring stops obeying Hooke's law.

Plan an experiment to do this. Your plan should be detailed enough for another student to follow.

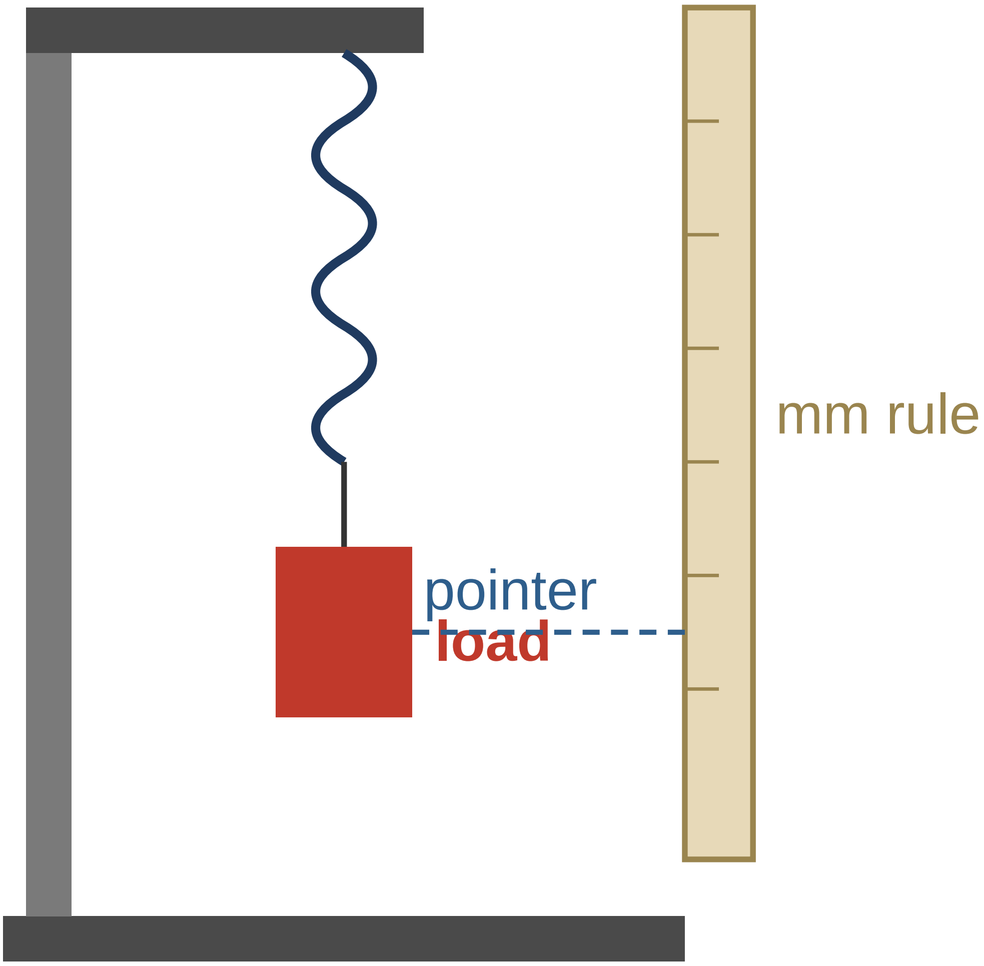

Fig. 7.1 — suggested apparatus (a guide only; the plan is yours to write).

In your plan, make sure you include:

i. the independent, dependent and controlled variables; [2]

ii. the apparatus and a step-by-step method, including how you measure the extension; [2]

iii. the readings you would take and how you would use them (what you would plot); [2]

iv. how the graph shows where the spring stops obeying Hooke's law; [1]

v. one safety precaution relevant to this experiment. [1]

Teach here: model the five-part skeleton aloud before the tutee writes. IV = load/force, DV = extension, controlled = same spring & temperature. Extension = new length − original (unstretched) length — not just the new reading. Plot extension vs load; straight-line-through-origin region obeys Hooke's law, the point where it curves is the limit. Safety: masses can fall → keep feet clear / put a tray or cushion below / wear goggles as spring may snap.

Question 8 ERRORS & EVALUATION DRILL **[7 marks]**

ACE

Short evaluation questions. These test the language examiners want in the "identify the error / suggest an improvement" parts.

1. A student times one swing of a pendulum with a stopwatch and gets 1.4 s. Explain why this single reading is unreliable and describe a better method. **[2]**
2. Distinguish between a *random* error and a *systematic* error, giving one example of each from a practical you have studied. **[2]**
3. A set of calipers reads 0.03 mm when the jaws are fully closed. Name this type of error and state exactly how the student should correct every reading. **[2]**
4. Explain why the improvement "repeat the experiment and take an average" does **not** help reduce a systematic error. **[1]**

Teach here: this is the highest-yield wrap-up. Hammer that improvements must be *specific and matched* to the named error. Zero error is systematic — subtracting the 0.03 mm fixes it; averaging never will because the offset is in every reading the same way.

Answer Key — Worked Marks

Mark points are shown as [1] style tags. Accept equivalent correct wording. Award method marks even if the final number is wrong (error carried forward).

Question 1 — Density of an irregular solid [7]

(a) Eye level with the bottom of the meniscus / water surface [1]; to avoid a parallax error when reading the scale [1].

(b) $V = V_2 - V_1 = 63.0 - 45.0 =$ **18.0 cm³** [1].

(c)

```
ρ = m / V = 47.25 / 18.0
  = 2.625 → 2.63 g/cm³
```

Correct substitution [1]; answer **2.63 g/cm³** to 3 s.f. with unit [1]. (3 s.f. because the raw data is given to 3 s.f.)

(d) Air bubbles cling to the stone / it is not fully submerged, so the displaced volume reads too high [1]; tap or tilt to release bubbles and make sure the stone is completely under the water [1].

Question 2 — Simple pendulum → g [11]

(a) Timing one swing has a large percentage (reaction-time) error [1]; timing 20 and dividing by 20 spreads that fixed error over many swings, reducing the uncertainty in T [1].

(b) Axes labelled with quantity *and* unit, sensible scale using > half the grid [1]; all five points plotted correctly as small crosses [1]; single thin best-fit line, roughly balanced above/below, passing near the origin [1].

(c) Using a large triangle on the line, e.g. from (0.20, 0.80) to (1.00, 4.02):

```
gradient = Δ(T²) / ΔL
         = (4.02 − 0.80) / (1.00 − 0.20)
         = 3.22 / 0.80
         ≈ 4.0 s²/m
```

Large triangle drawn / values read off the line [1]; gradient ≈ **4.0 s²/m** (accept 3.9–4.1) [1].

(d) gradient = $4\pi^2/g$, so

```
g = 4π² / gradient
  = 39.48 / 4.0
  ≈ 9.9 m/s²
```

Correct rearrangement [1]; g ≈ **9.9 m/s²** (accept 9.6–10.1) [1].

(e) Any one [1]: reaction-time error in timing; amplitude of swing too large; air resistance / draughts; string not measured to the centre of the bob.

Question 3 — Principle of moments [8]

(a) For a body in equilibrium, the sum of clockwise moments about a pivot equals the sum of anticlockwise moments about the same pivot [1].

(b) 100 g mass: 50.0 − 15.0 = 35.0 cm from pivot; mass M: 80.0 − 50.0 = 30.0 cm from pivot [1].

(c) Taking moments about the pivot (g cancels, so use mass × distance):

```
100 × 35.0 = M × 30.0
  3500 = 30.0 M
  M = 3500 / 30.0 = 116.7 g
```

Correct moment equation [1]; rearrangement [1]; **M ≈ 117 g** (accept 116–117 g) [1].

(d) The rule is uniform, so its weight acts at its centre of gravity, which is at the pivot; its perpendicular distance from the pivot is zero, so its moment is zero [1].

(e) Any one precaution with reason [2]: use a knife-edge / sharp pivot so the balance point is well defined; check the rule is horizontal (e.g. with a set square) so the perpendicular distances equal the marked distances; take the balance point as the midpoint of the range where it just tips either way.

Question 4 — Resistance of a wire [12]

(a) R = V/I for each row:

```
L=20.0: 0.42/0.60 = 0.70 Ω
L=40.0: 0.84/0.60 = 1.40 Ω
L=60.0: 1.26/0.60 = 2.10 Ω
L=80.0: 1.68/0.60 = 2.80 Ω
L=100.0: 2.10/0.60 = 3.50 Ω
```

All values correct [1]; consistent 2–3 s.f. with unit Ω in heading [1].

(b) R is directly proportional to L ($R \propto L$) [1]; e.g. when L doubles from 20 to 40 cm, R doubles from 0.70 to 1.40 Ω — the ratio R/L is constant [1].

(c) Straight line [1] passing through the origin, with axes labelled R / Ω (y) against L / cm (x) [1].

(d) Current heats the wire; a hotter metal wire has a higher resistance [1]; switching off between readings keeps the wire near room temperature so R depends only on length (avoids a systematic error) [1].

(e) She adjusted the rheostat / variable resistor after changing the length to bring the current back to 0.60 A [1]; component = rheostat (variable resistor) [1].

(f) Any one [2]: end of wire not exactly at the zero mark / difficulty seeing the contact point → view square-on to avoid parallax, use a set square, or take the length from clearly fixed contact points; wire not straight → keep it taut along the rule.

Question 5 — Refraction through a glass block [9]

(a) Place both pins vertically (upright) [1] and as far apart as possible along the incident ray, so the line joining them is long and the direction is defined accurately [1].

(b) Light slows down on entering the denser glass [1]; the part of the wavefront that enters first slows first, so the ray bends towards the normal [1].

(c)

```
n = sin i / sin r
  = sin 50° / sin 30°
  = 0.766 / 0.500
  = 1.53
```

Correct formula & substitution [1]; **n ≈ 1.53** [1].

(d) 3 s.f. (or 2 s.f.), because the sine values / measured angles are known to about that precision [1].

(e) Any two [2]: use a sharp pencil and measure angles from the normal (not the surface); use long ray lines so the protractor can be read accurately; use pins far apart; repeat for several angles of incidence and take an average / plot sin i against sin r.

Question 6 — Cooling curve & heat loss [10]

(a) About **55 °C** (accept 52–58 °C read from the curve at t = 4.0 min) [1].

(b) The rate of cooling is fast at first and slows down / the curve gets less steep [1]; because the rate of heat loss depends on the temperature difference between the water and the surroundings [1]; as the water cools this difference shrinks, so energy is transferred more slowly [1].

(c) Any two changes, each with a reason [4]: add a lid → reduces heat loss by evaporation/convection from the top; wrap the beaker in insulation/lagging → reduces conduction and convection from the sides; stand it on an insulating mat → reduces conduction through the base; use a smaller surface area / cover shiny surface → reduces radiation.

(d) Cooling stops when the water reaches the temperature of its surroundings (room temperature) [1]; there is then no temperature difference, so no *net* transfer of energy to the surroundings [1].

Question 7 — Planning task (Hooke's law) [8]

(i) Variables [2]: independent = load / force applied; dependent = extension of the spring; controlled = same spring, same starting (unstretched) length, constant temperature.

(ii) Apparatus & method [2]: clamp the spring vertically from a stand; fix a metre rule alongside with a pointer at the spring's lower end; record the unstretched pointer reading; add a known mass, wait for the spring to settle, record the new reading; extension = new reading − original reading; repeat adding masses in equal steps.

(iii) Readings & graph [2]: take 6–8 loads over a sensible range; record load and extension in a table; plot extension (y) against load (x).

(iv) Interpreting the graph [1]: while the spring obeys Hooke's law the graph is a straight line through the origin; the load at which the line starts to curve (bend away from straight) is where it stops obeying Hooke's law (the limit of proportionality).

(v) Safety [1]: masses may fall — keep feet clear / place a tray or cushion beneath; or wear eye protection in case the overstretched spring snaps.

Award full marks only if the method is genuinely repeatable by another student and the fair-test controls are stated.

Question 8 — Errors & evaluation drill [7]

(a) One swing is comparable to the ~0.2 s reaction time, so the percentage error is very large [1]; time many oscillations (e.g. 20) and divide by the number, ideally starting/stopping at a fiducial mark as the bob passes the centre [1].

(b) Random error: scatters readings both sides of the true value (e.g. reaction-time timing) — reduced by repeating and averaging [1]. Systematic error: shifts every reading the same way (e.g. zero error on an instrument, heat loss) — reduced by fixing the cause [1].

(c) A zero error (systematic) [1]; subtract 0.03 mm from every reading taken with the calipers [1].

(d) Because the same offset is present in every reading, so averaging repeated readings keeps the offset — it does not cancel out; only correcting the cause removes it [1].

Closing checklist to run with your tutee

Units in every column heading and final answer • consistent significant figures • graph fills > half the grid with sensible scales • gradient triangle spans > half the line • every "improvement" is specific and matched to a named error. Nail these five and the technique marks look after themselves.

Written to the SEAB Singapore–Cambridge O-Level Physics syllabus 6091 (first examined 2026). Questions, data and figures are original teaching material, not reproductions of past papers. From 2027 the O- and N-Level examinations consolidate into the Singapore–Cambridge Secondary Education Certificate (SEC); confirm practical requirements against current SEAB documents for later cohorts.